Syed Ishrak Zaman

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Education

University of Massachusetts Amherst Amherst, MA
Bachelor of Science in Computer Engineering – GPA: 3.5/4.0 *May 2026*
- **Relevant Coursework:** Machine Learning, Artificial Intelligence, Embedded Systems, Systems Programming, Probability & Statistics, Linear Algebra
- **Honors:** Dean's List (2023, 2024)

Technical Skills

Languages: Python, C, C++, MATLAB, Bash, ARM Assembly

ML/AI Frameworks: PyTorch, TensorFlow, Hugging Face, OpenCV, scikit-learn, Pandas, NumPy

ML/AI Concepts: Computer Vision, Deep Learning, Object Detection (YOLO), CLIP Embeddings, FAISS, OCR, Transfer Learning, Model Evaluation, Data Preprocessing

APIs & Tools: OpenAI API, GitHub, Jupyter, VS Code

Hardware/Embedded: Raspberry Pi 5, Arduino, BeagleBone Black, nRF Connect SDK, KiCad

Projects

Smart Grocery Checkout System – Python, YOLOv8, CLIP, FAISS, FastAPI, Raspberry Pi 5 Fall 2025 – May 2026
- Developed an autonomous checkout system combining YOLOv8 object detection, CLIP visual embeddings, and FAISS vector search to identify grocery items without barcodes, achieving 100% Top-1 classification accuracy across 16 SKUs with zero misclassifications.
- Built a preprocessing and dataset generation pipeline (GroZi-120) to populate a SQLite product database of 1,007 SKUs; end-to-end checkout completes in under 6 seconds with receipt emailed automatically via SMTP.
- Integrated dual cameras, 5 ultrasonic sensors, a conveyor belt, and a Raspberry Pi 5 into a full hardware-software pipeline as Computer Vision & AI Lead; system ran continuously for 12+ hours with zero crashes across 100 sensor trials.

Compact Temperature Monitoring Device – Embedded C, KiCad, ATtiny85 Spring 2025
- Designed and fabricated a compact PCB (1.34 in x 1.49 in) in KiCad using an ATtiny85 MCU, TMP-36 analog sensor, and SSD1306 OLED display.
- Programmed firmware in Embedded C for real-time temperature sampling with 0.1°C resolution and threshold-based visual alerts on the OLED display.

5G IoT Timing Analysis – nRF Connect SDK, nRF9160DK, Serial Logging Fall 2024
- Investigated timing and synchronization discrepancies in 5G IoT communication pipelines using the nRF9160DK and iBasis IoT SIM; characterized latency patterns via serial logging and nRF Connect SDK.

Experience

Scale AI Remote
AI Model Trainer *February 2024 – June 2024*
- Evaluated and refined AI-generated code across Python, C, and Assembly to improve correctness, consistency, and alignment with programming task prompts.
- Contributed to RLHF pipelines for large language models, focusing on code generation accuracy and reasoning quality.

University of Massachusetts Amherst Amherst, MA

Undergraduate Teaching Assistant, Intro Python (ECE) *January 2025 – May 2026*
* Designed grading rubrics and automated pytest suites for Python projects covering OOP, data structures, and algorithm design.
* Held office hours and guided 100+ students through debugging, code review, and conceptual problem-solving.

Undergraduate Teaching Assistant, Intro to Security Engineering *January 2025 – May 2026*
* Designed and graded 2 hands-on labs covering RSA & DES encryption and PRNG/TRNG random number generation.

Resident Assistant *August 2025 – May 2026*
* Mentor 40+ residents; facilitate conflict resolution, conduct monthly well-being check-ins, and organize community programming events.

Security Monitor *August 2023 – May 2025*
* Maintained building safety through access control, incident documentation, and university policy enforcement.